Xiaohan Fei

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Education

University of California, Los Angeles — Sept. 2014 - Sept. 2019
Ph.D. in Computer Science
Research Group: UCLA Vision Lab (http://vision.ucla.edu)
GPA: 3.88/4.0
Thesis: Inertial-aided Visual Perception of Geometry and Semantics
Advisor: Prof. Stefano Soatto

Zhejiang University, Hangzhou, China — Sept. 2010 - June 2014
B.Eng. in Information and Communication Engineering
Minor: Advanced honor Class of Engineering Education (ACEE), Chu-Kechen College
GPA: 3.98/4.0(92.35/100)
Thesis: Wide-baseline feature matching for panoramic images
Thesis advisor: Prof. Zhiyu Xiang

Professional Interests

I have strong fundamentals in computer vision, robotics, and machine learning, and proven ability to quickly gain knowledge and contribute to newer and emerging areas. I've led multiple zero to one projects in the AI/ML industry including the localization and mapping initiative at AWS, the video generation foundation model at Amazon AGI, and my pioneering work on using multi-modal data for depth perception that won me a **Best Paper Award** in Robot Vision at ICRA 2019.

Open-source Software

XIVO: a state-of-the-art localization and mapping software, forked by 100+ and starred by 800+ developers worldwide. For the rest, see my github page: https://github.com/feixh

Work Experience

Amazon AGI, Bellevue, Washington — January 2026 - Present
Principal Applied Scientist
AWS Agentic AI

Amazon AGI, Bellevue, Washington — December 2023 - December 2025
Principal Applied Scientist
Principal tech lead of Amazon Nova Reel – Amazon's video generation foundation model. I led both 1/ the technical development with hands-on experiences working with my science and engineering team on a daily basis, and 2/ strategic direction of Amazon Nova Reel. The model was launched at AWS re:Invent 2024.

AWS AI Labs, Bellevue, Washington — April 2020 - December 2023
Senior Applied Scientist
Tech lead of a multi-sensor localization and mapping initiative at Amazon. The tech stack 1/ supports a wide range of sensors including IMU, camera (stereo and mono, different lens types), and wheel encoder, 2/ runs on both cloud and edge devices, and 4/ was adopted by multiple Amazon teams ranging from AWS AI Labs to Amazon Ring.

Facebook Reality Labs, Redmond, Washington — September 2019 - April 2020
Senior Research Scientist
Surreal team, multi-sensor localization and mapping for AR/VR.

Awards & Distinctions

2019: **Best Paper Award** in Robot Vision, out of 2900 submissions, at ICRA 2019
2013: **Meritorious Winner** of Mathematical Contest in Modeling (top 15% of 6000 teams worldwide)
2012: **National Scholarship** (the highest honor for undergraduates in China)

Publications

* indicates equal contribution

[1] Sirui Xu, Samuel Schulter, Morteza Ziyadi, Xialin He, **Xiaohan Fei**, Yu-Xiong Wang, Liang-Yan Gui. InterPrior: Scaling Generative Control for Physics-Based Human-Object Interactions. CVPR, 2026.

[2] Amazon Nova 2: Multimodal Reasoning and Generation Models. Tech Report. 2025

[3] The Amazon Nova Family of Models: Technical Report and Model Card. Tech Report. 2024.

[4] Chethan Parameshwara*, Alessandro Achille*, Matthew Trager, Xiaolong Li, Jiawei Mo, Ashwin Swaminathan, CJ Taylor, Dheera Venkatraman, **Xiaohan Fei***, Stefano Soatto*. Towards visual foundational models of physical scenes. In *arXiv*, 2023.

[5] **Xiaohan Fei**, Chethan Parameshwara, Jiawei Mo, Xiaolong Li, Ashwin Swaminathan, CJ Taylor, Paolo Favaro, Stefano Soatto. A Quantitative Evaluation of Score Distillation Sampling Based Text-to-3D. In *arXiv*, 2023.

[6] Ziqi Lu, Jianbo Ye, **Xiaohan Fei**, Xiaolong Li, Jiawei Mo, Ashwin Swaminathan, Stefano Soatto. Fast sparse view guided nerf update for object reconfigurations. In *arXiv*, 2023.

[7] Xiaolong Li, Jiawei Mo, Ying Wang, Chethan Parameshwara, **Xiaohan Fei**, Ashwin Swaminathan, CJ Taylor, Zhuowen Tu, Paolo Favaro, Stefano Soatto. Grounded compositional and diverse text-to-3d with pretrained multi-view diffusion model. In *arXiv*, 2023.

[8] **X. Fei**, H. Wang, X. Zeng, L. Cheong, J. Tighe. Single View Physical Distance Estimation using Human Pose. In *International Conference on Computer Vision* (ICCV), 2021.

[9] A. Wong, **X. Fei**, B. Hong, and S. Soatto. An Adaptive Framework For Learning Unsupervised Depth Completion. In *IEEE Robotics and Automation Letters* (RA-L), 2021.

[10] A. Wong*, **X. Fei***, and S. Soatto. Unsupervised Depth Completion from Visual-Inertial Odometry. In *International Conference on Robotics and Automation* (ICRA), 2020. Also in *IEEE Robotics and Automation Letters* (RA-L).

[11] **X. Fei**, A. Wong, and S. Soatto. Geo-Supervised Visual Depth Prediction. In *International Conference on Robotics and Automation* (ICRA), 2019. (**Best Paper Award in Robot Vision**) Also in *IEEE Robotics and Automation Letters* (RA-L).

[12] **X. Fei**, S. Soatto. Visual-Inertial Object Detection and Mapping. In *European Conference on Computer Vision* (ECCV), 2018.

[13] J. Dong*, **X. Fei***, and S. Soatto. Visual-Inertial-Semantic Scene Representation for Object Detection. In *Computer Vision and Pattern Recognition* (CVPR), 2017.

[14] **X. Fei**, K. Tsotsos, and S. Soatto. A Simple Hierarchical Pooling Data Structure for Loop Closure. In *European Conference on Computer Vision* (ECCV), 2016.

Professional Services

Reviewer of top computer vision (CVPR, ICCV, ECCV), robotics (ICRA, IROS), and artificial intelligence (NeurIPS, ICLR, AAAI) conferences.

Relevant Coursework

University of California, Los Angeles: Machine Perception (Prof. S. Soatto), Convex Optimization (Prof. L. Vandenberghe), Calculus of Variations (Prof. L. Vese), Vision as Bayesian Inference (Prof. A. Yuille), Applied Probability (Prof. Y. Wu), Theoretical Statistics (Prof. A. Amini), Numerical Analysis (Prof. J. Teran), Machine Learning Algorithm (Prof. M. Sarrafzadeh)
Zhejiang University: Computer Vision (Pof. Z. Xiang), Spectral Analysis of Signals (Prof. X. Gong), Information Theory (Prof. Z. Zhang), Mathematical Modeling (Prof. Q. Yang)

Relevant Skills

Programming Language: Python, C++
Software Framework: Deep Learning (PyTorch, Megatron, TensorFlow), Vision and Robotics (OpenCV, ROS), Math & Optimization (Eigen, Ceres)